Exploring the Impact of Encouragement and Student Behaviour on Academic Effort in Second-year Statistics Courses*

Motivational Emails Boost Effort, Perception Drive Effort, Difficulty Reduces Effort

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Student engagement is a key factor in academic success. This study uses Bayesian multiple linear regression to explore how emotional states, cognitive competence, task difficulty, intrinsic interest, and email intervention types collectively shape student effort in email-based interventions. The findings reveal that emotional and motivational factors, especially changes in affect and interest, strongly enhance effort, while task difficulty requires careful balance to avoid demotivation. These results offer practical guidance for developing cost-effective, personalized interventions that integrate emotional and cognitive support to boost academic motivation.

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*Code and data are available at: https://github.com/Sophiaaa-Y/student-effort-analysis.

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1 Introduction

Student engagement plays an important role in shaping academic outcomes, acting as the driving force behind students' ability to stay motivated, overcome challenges, and achieve success in their educational pursuits (Kahu and Nelson 2018). While research has explored its emotional, cognitive, and behavioral dimensions (Goes, Guo, and Lin 2016), less is known about their combined effects within targeted interventions. This paper examines how email-based academic interventions influence effort through emotional states, cognitive competence, task difficulty, and intrinsic interest (Wang and Degol 2014).

This study explores the factors influencing changes in students' study effort in response to an email-based academic intervention designed to deliver motivational messages aimed at fostering interest in statistics and encouraging active learning in the subject. By analyzing variables such as emotional states, cognitive competence, perceived difficulty, intrinsic motivation, and email type, the research aims to uncover how these predictors collectively impact variations

in effort. Using a Bayesian multiple linear regression model implemented with the rstanarm package in R (Goodrich et al. 2022), the analysis draws on over 600 student records to explore the combined effects of these factors on academic engagement. Unlike prior studies that examined these predictors individually (Mac Iver, Stipek, and Daniels 1991) (Chevalier, Dolton, and Lührmann 2014), this research offers a holistic framework for understanding how various aspects of student engagement and communication strategies interact to drive behavioral changes in academic contexts.

Our findings shows a deliberate focus on increasing student interest and confidence to enhance academic effort. The analysis emphasizes that interventions promoting positive emotions and cognitive competence were most effective, while moderate task difficulty encouraged persistence. Strategic use of engaging email types further amplified these effects, boosting motivation and improving overall student engagement.

The significance of these insights also offer practical guidance for designing interventions that effectively enhance academic engagement and motivation. By integrating emotional and cognitive support with personalized communication, these findings provide a foundation for cost-effective, evidence-based strategies in education. Furthermore, this study contributes to the broader discourse on addressing systemic challenges in student engagement, enriching discussions in educational psychology and behavioral science, as explored by scholars like Pajares (2001) and Winne and Nesbit (2010).

This paper is structured as follows. The Section 1 includes the introduction part of this study, and the Section 2 outlines the data that we used in the study, and describes the sources of the dataset, the behavioral and performance variables selected for analysis, and the reason reasons behind the choice. Section 3 introduces the Bayesian multiple linear regression model that we used to analyze the relationships between students' effort and predictors such as emotional shifts, cognitive competence, difficulty perception, intrinsic interest, and email type, which provide a statistical framework to understand how they influence students' effort. In Section 4, it shows the findings from the Bayesian multiple linear regression model, including statistical summaries and graphical representations of the model's outcomes, which show details about how different factors contributed positively or negatively to variations in effort. Besides, Section 5 interprets these results in the context of email interventions, discusses limitations of the study, and suggests directions for future research to enhance understanding of student engagement. Finally, additional details about the dataset and its processing are included in Section A, Section B includes diagnostic plots and posterior predictive checks to evaluate the model's reliability, and Section C provides an in-depth exploration of survey and observational data methodology, linking to relevant literature and including simulations to contextualize and validate the study's approach.

2 Data

2.1 Data Overview

The dataset, sourced from the Nudge Data Repository (Taback and Gibbs 2022), originates from a randomized experiment on students in the "Practice of Statistics I" (STA220H1F) course at the University of Toronto (Taback and Gibbs 2023). The experiment examined the effects of flipped and fully online sections, as well as weekly email interventions, on academic performance and attitudes toward statistics. Variables include demographic details such as section enrollment, program type, academic performance such as quiz scores, final exam results, and behavioral metrics such as participation in flipped or online sections. Surveys measured attitudes before and after the course, assessing interest, confidence, and perceived value. This dataset provides a robust framework for analyzing how instructional methods and interventions influence student outcomes and attitudes.

2.2 Measurement

The dataset used in this study was sourced from the Nudge Data Repository (Taback and Gibbs 2022). It reflects real-world educational phenomena, examining how email interventions influence student attitudes toward statistic. Each observation corresponds to a unique student who received an intervention, with data collected on their behaviors and performance metrics before and after the emails. According to Taback and Gibbs (2023), the measurements in this study were carefully designed to align with established behavioral and educational theories, ensuring reliability and validity. Self-reported surveys were used to capture subjective variables, such as changes in emotional states, cognitive competence, and intrinsic motivation. These surveys employed structured Likert-scale items, ranging from "strongly disagree" to "strongly agree," allowing participants to express nuanced levels of agreement or disagreement. Emotional states, such as interest and frustration, were assessed before and after the intervention to evaluate the effectiveness of motivational strategies, while cognitive competence was measured by asking participants to rate their confidence in handling statistical tasks. Objective metrics complemented these surveys by providing unbiased data on task complexity and performance outcomes. Task complexity was quantified based on the number of steps required to complete problems and the time spent on each task, while performance outcomes were measured against standardized benchmarks, such as quiz scores and task completion rates. This dual-method approach ensured a comprehensive understanding of how interventions influenced behavior, capturing both psychological engagement and actual performance. Grounded in theoretical models like self-determination theory and cognitive load theory, the study integrated subjective and objective measures to provide a robust framework for analyzing the predictors of academic effort and engagement.

This dataset originally has 1430 records. To ensure the reliability of the analysis, the dataset was pre-processed to retain only 622 rows where:

1. The email intervention was opened.
2. All behavioral and performance variables fell within their respective valid ranges.
3. Observations were complete, with no missing values.

The data for this study was systematically downloaded, cleaned, analyzed, modeled and visualized using R (R Core Team 2023), a extensive statistical programming language. The following packages were used for this study:

- **`tidyverse`** (Wickham et al. 2019): Streamlined data manipulation and visualization workflows enabled efficient filtering, summarizing, and transforming of large datasets while providing flexible and detailed visualizations for both exploratory and inferential analysis, showcasing variable distributions and relationships.
- **`janitor`** (Firke 2021): Streamlining the management of raw data by standardizing variable names and organizing data structures for easier analysis.
- **`arrow`** (Apache Arrow 2021): Optimized for rapid reading and writing of large datasets, improving data management efficiency.
- **`knitr`** (Xie 2021): Allowed for dynamic integration of code, analysis, and visualizations into a cohesive and reproducible report.
- **`bayesplot`** (Gelman, Gabry, et al. 2021): Used to generate visualizations for posterior predictive checks and diagnostics, ensuring thorough evaluation of the model's performance.
- **`rstanarm`** (Goodrich et al. 2022): Enabled seamless implementation of Bayesian regression models, offering an accessible interface for utilizing Stan's capabilities.
- ***Telling Stories with Data*** (Alexander 2023): This book served as a reference for its techniques and methodologies in effectively presenting data and statistical insights.

Table 1: Preview of Key Factors Influencing Student Effort

email	change in affect	change in cognitive competence	change in difficulty	change in interest	change in effort
Interesting	-1.50	-2.67	-1.43	-1.00	-2.50
Interesting	0.33	0.33	-0.29	0.50	-0.75
Plain	-2.17	-1.33	-0.14	-0.25	-1.50
Interesting	0.33	0.33	0.29	-0.25	-0.25
Interesting	0.67	0.33	0.14	-0.75	-2.75
Interesting	0.67	0.17	0.71	-1.50	0.00

Table 1 presents the first six rows from the cleaned dataset, focusing on key factors that may influence a student's effort.

2.3 Variables

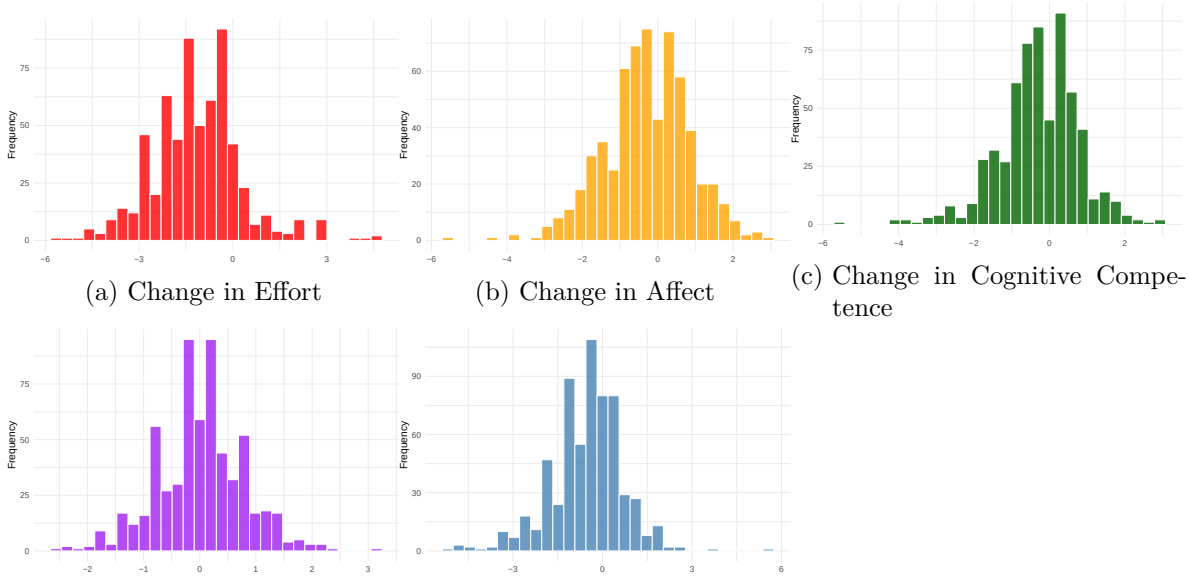
This analysis focuses on the following variables, with a primary interest in "change in effort" as the dependent variable. And detailed information about these variables is shown in Table 1, which shows the first few records of the dataset.

2.3.1 Outcome Variable

- **change in effort:** Measures variations in the amount of effort students report putting into their academic tasks pre- and post-intervention. It reflects how the interventions influence students' willingness to invest time and energy into their work. An increase indicates heightened persistence and engagement, while a decrease suggests diminished motivation.

2.3.2 Predictor Variables

- **change in affect:** Change in students' feelings concerning to the course material before and after the intervention. Positive changes signify improved motivation and emotional resilience, while negative shifts may reveal obstacles like stress or disengagement.
- **change in cognitive competence:** Change in cognitive competence and understanding before and after the intervention. Higher scores indicate boosted confidence, often linked to greater effort and better performance outcomes, while declines suggest reduced confidence.
- **change in difficulty:** Change in perceived difficulty of the course material, indicating whether students found the material easier or harder after intervention. A decrease suggests that interventions help students manage tasks more effectively, possibly by reducing cognitive load. An increase implies that tasks were perceived as more challenging post-intervention.
- **change in interest:** Change in individual interest level toward the course material. Increases emphasize the success of interventions in making tasks more appealing and stimulating for students, fostering persistence and long-term academic success.
- **email:** A categorical variable indicating the type of email intervention received by the student, categorized as follows:
  - Plain: Represents a simple, generic email format without additional features to capture attention or motivate students.
  - Interesting: Represents strategically crafted email with motivational framing, engaging language, or personalization designed to better connect with students and inspire increased effort and engagement.



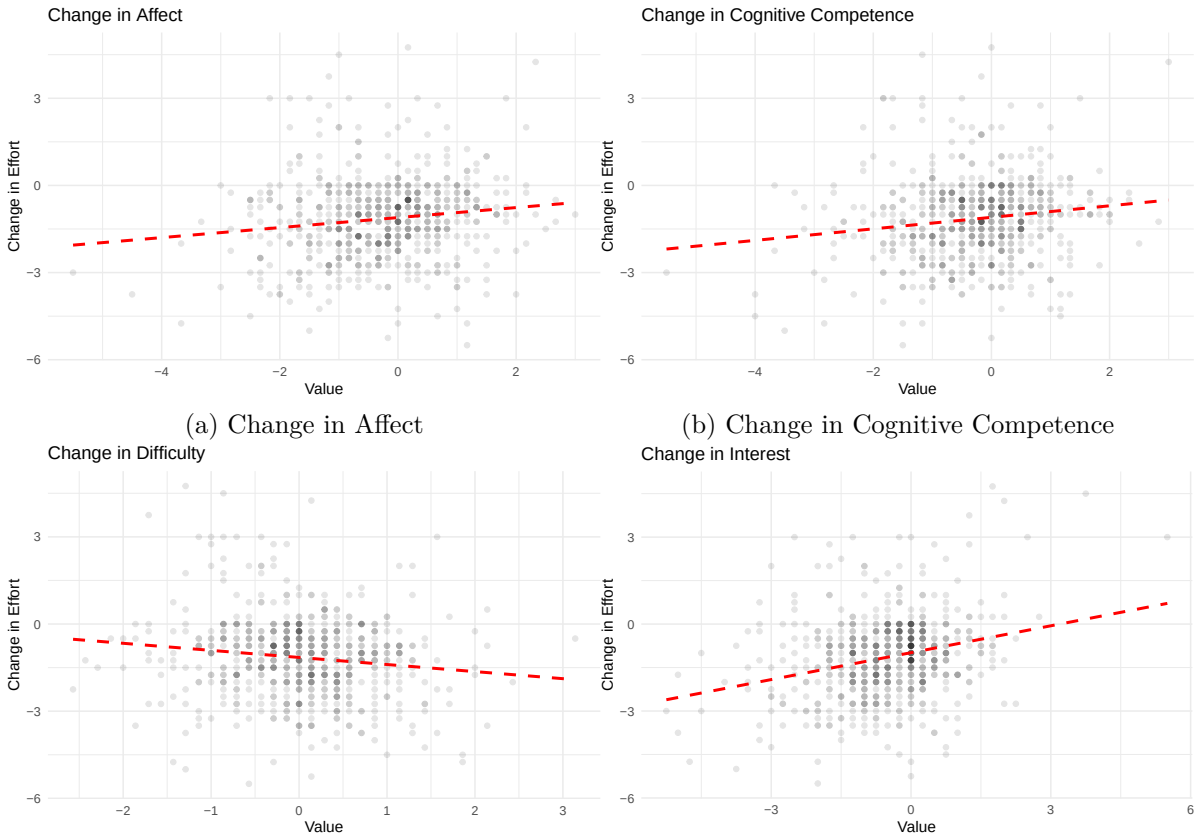
(a) Change in Effort (b) Change in Affect (c) Change in Cognitive Competence

(d) Change in Difficulty (e) Change in Interest

Figure 1: Distributions of Changes in Behavioral and Performance Variables: Effort shows a slightly skewed distribution, Affect and Interest exhibit broad normal-like spreads, Cognitive Competence is narrowly distributed, and Difficulty shows a slight negative skew

Figure 1 provides a detailed view of the processed data, illustrating the distributions of key behavioral and performance variables captured in the study. The facet plot highlights patterns in variables such as change in effort, change in affect, change in cognitive competence, change in difficulty, and change in interest. Most behavioral variables show distributions centered around zero, indicating that, on average, changes in effort, affect, cognition competence, difficulty, and interest were minimal.

From the visualization, we can discern slight skews in specific variables. For instance, the change in effort shows slightly negative, suggesting a moderate decline in effort across the sample, while the change in interest shows a mild negative skew, indicating a more frequent decrease in interest. These distributions provide valuable advises for student responses to the intervention and their variability.



(a) Change in Affect

(b) Change in Cognitive Competence

(c) Change in Difficulty

(d) Change in Interest

Figure 2: Relationships Between Change in Effort and Predictors: Affect and Interest Show Positive Trends with Broad Variability, Cognitive Competence Exhibits a Strong Positive Linear Relationship, and Difficulty Displays a Negative Trend

Figure 2 illustrates the relationships between change in effort and its predictors such as change in affect, change in cognitive competence, change in difficulty, and change in interest, using scatter plots with regression lines for each visualization. Each represents a unique predictor, allowing for an individual examination of their influence on changes in student effort.

Based on the regression lines, the visualization provides insight into the trends within the data. For the change in affect and change in interest exhibit positive associations with change in effort, suggesting that as students experience positive shifts in affect and interest, their effort tends to increase. However, change in difficulty appears to have a weak or slightly negative relationship, indicating that perceived difficulty may slightly decrease effort. These visualizations provide a comprehensive overview of how these predictors contribute to changes in effort, guiding further analysis to confirm their statistical significance.

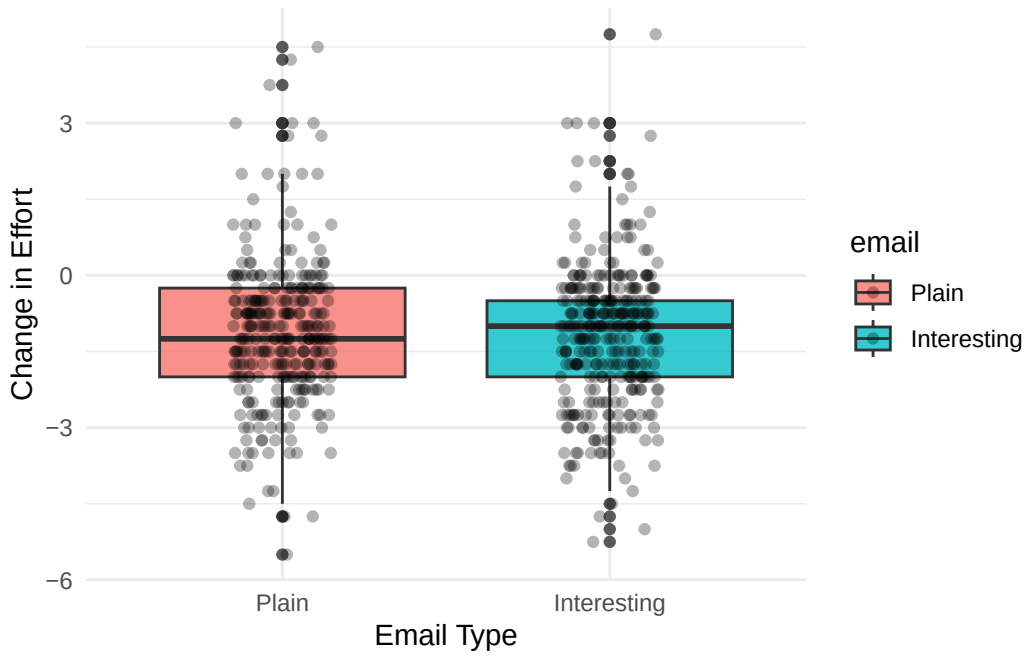


Figure 3: Variations in Change in Effort by Email Type: Boxplot and Jitter Plot Highlighting Differences in Median, Variability, and Outliers Between Plain and Interesting Emails

Figure 3 illustrates the distribution of changes in effort among students based on two types of email interventions: "Plain" and "Interesting." Both groups show a median change in effort around 0, indicating that, on average, the emails did not substantially affect effort levels for most students. However, the interquartile range (IQR), representing the middle 50% of the data, is slightly wider for the "Interesting" email group, suggesting greater variability in responses to this intervention. Additionally, the "Interesting" group exhibits several extreme negative outliers, with changes below -3, indicating that a subset of students experienced a notable decrease in effort. In contrast, the "Plain" group has fewer and less extreme outliers, showing a more consistent response pattern. The broader spread and more pronounced outliers in the "Interesting" group suggest that while these emails may have a greater impact on some students, the responses are highly variable. This highlights the importance of exploring additional factors, such as student characteristics or baseline engagement, to better understand the varying effects of different email interventions.

2.4 Justification of Variable Choice

The variables chosen for this study were carefully selected based on their relevance to understanding the factors influencing changes in student effort, as captured in the dataset from the Nudge Data Repository (Taback and Gibbs 2022). Each variable provides critical insights into different dimensions of behavioral and academic outcomes:

- **change in affect:** Tracks shifts in students' emotional states pre-intervention and post-intervention, showing how interventions influence feelings about academic tasks. Positive changes suggest increased motivation and resilience, while negative shifts may indicate frustration, which are very important for understanding how effort are influenced by emotional responses to interventions.
- **change in cognitive competence:** This means change in cognitive competence, which measures changes in students' perceived learning ability, providing insights into confidence and self-efficacy. Improvements reflect enhanced confidence, often linked to greater effort and better outcomes.
- **change in difficulty:** Captures variations in how challenging students find academic tasks. Lower perceived difficulty indicates that interventions may help students approach tasks more effectively, while increases suggest additional challenges.
- **change in interest:** Assesses shifts in intrinsic motivation and engagement, critical for fostering long-term academic persistence. Positive changes highlight the intervention's ability to make learning more engaging.
- **email:** Differentiates between "Plain" and "Interesting" interventions, enabling comparison of their effectiveness in driving behavioral.

Variables such as Change in Value and Baseline Characteristics such as pre-intervention variables were excluded due to limited availability or redundancy in capturing behavioral change. Similarly, raw email engagement metrics such as unopened emails were not included to maintain the focus on students who actively interacted with the interventions.

Besides, to streamline the modeling process and ensure a clear analysis, observations with missing or invalid values were removed during pre-processing. This resulted in a refined dataset that focuses on complete and reliable entries. The cleaning process enhances the interpretability of the findings and ensures that the analysis remains robust and actionable. This careful selection and preparation of variables allow the study to comprehensively examine the relationships between effort and its potential predictors, providing meaningful insights into the effectiveness of behavioral nudges in an educational context.

3 Model

This study employs a Bayesian multiple linear regression model to analyze the factors influencing changes in student effort following an email-based intervention.

The model integrates several key predictors:

- Change in Affect: Measures shifts in emotional states as a response to the intervention.
- Change in Cognitive Competence: Reflects variations in students' perceived learning ability and confidence.

- Change in Difficulty Perception: Measures changes in how students perceive the challenge of academic tasks.
- Change in Interest: Represents shifts in intrinsic motivation toward academic tasks.
- Email Type: Differentiates between "Plain" and "Interesting" interventions to assess the impact of communication style.

The model estimates the relationship between these predictors and the outcome variable, change_effort, which quantifies how each factor contributes to the variations in student effort. By employing the `rstanarm` package (Goodrich et al. 2022), the Bayesian approach accommodates uncertainty in the coefficients and provides posterior distributions for each parameter.

Background details and diagnostics are included in Appendix B.

3.1 Model set-up

Let y_i represent the continuous outcome variable representing the change in effort, for the i-th student, capturing the variation in effort pre-intervention and post-intervention. The predictors in the Bayesian multiple linear regression model include:

- β_0: The intercept, representing the expected value of "change in effort" when all predictor variables are at their reference or baseline values.
- β_1: The coefficient for "change in affect", measuring shifts in emotional states as a response to the intervention. Higher values indicate more positive emotional engagement changes.
- β_2: The coefficient for "change in cognitive competence", reflecting variations in perceived learning ability. This predictor provides insights into changes in students' confidence and cognitive self-assessment.
- β_3: The coefficient for "change in difficulty", capturing changes in students' perception of the challenge level of academic tasks. Lower values signify reduced perceived difficulty.
- β_4: The coefficient for "change in interest", representing shifts in intrinsic motivation and engagement with academic tasks. Higher values indicate increased interest.
- β_5: The coefficient for the categorical variable "email", differentiating the intervention types ("Plain" vs. "Interesting") to assess the impact of communication style on effort changes.

Each coefficient β_j quantifies the contribution of the j-th predictor to the outcome variable, "change in effort".

- η_i: The linear predictor for the i-th observation. It combines the intercept β_0 and coefficients multiplied by the predictor variables, representing the expected value of the outcome variable.
- ϵ_i: The residual error for the i-th observation, assumed to follow a normal distribution with mean 0 and variance σ^2.

$$
\begin{aligned}
y_i &\sim \text{Normal}(\eta_i, \sigma^2) && (1)\\
\eta_i &= \beta_0 + \beta_1 \times \text{change in affect}_i + \beta_2 \times \text{change in cognitive competence}_i && (2)\\
&\quad + \beta_3 \times \text{change in difficulty}_i \ + \beta_4 \times \text{change in interest}_i \ + \beta_5 \times \text{email}_i && (3)\\
\beta_j &\sim \text{Normal}(0, 2.5) \text{ (default weakly informative prior)} && (4)\\
\sigma^2 &\sim \text{Exponential}(1) \text{ (default prior for error variance)} && (5)
\end{aligned}
$$

3.2 Prior distributions

In the Bayesian multiple linear regression model implemented using the `rstanarm` package (Goodrich et al. 2022), default priors from `rstanarm` (Goodrich et al. 2022) are applied to ensure robust and reliable inference. These priors are weakly informative, striking a balance between regularization and adaptability to the data:

- **Intercept Priors**: The model's intercept is assigned a normal prior distribution with a mean of 0 and a standard deviation of 2.5. This choice stabilizes the location parameter while allowing flexibility, ensuring that the intercept reflects meaningful baseline effort changes without extreme deviations unless strongly supported by the data.
- **Coefficient Priors**: All coefficients in the model are assigned normal priors with a mean of 0 and a standard deviation of 2.5. This prior setup provides weakly informative regularization, constraining the range of plausible values for predictors such as change in affect, change in cognitive competence, change in difficulty, change in interest, and email. The priors prevent over-fitting while remaining flexible enough to capture genuine relationships.
- **Error Variance Priors**: The variance of the residuals (σ^2) follows an exponential prior with a rate parameter of 1. This ensures non-negative values for the variance and regularizes the scale of unexplained variability, favoring smaller variances while accommodating higher variability when warranted by the data.

We run the model in R (R Core Team 2023) using the `rstanarm` package (Goodrich et al. 2022). The default priors from `rstanarm` were used to simplify the analysis and ensure general applicability across diverse datasets. These priors enhance the model's interpret-ability and stability, providing meaningful inferences about the factors influencing changes in student effort while retaining flexibility to adapt to the nuances of the observed data.

4 Results

The analysis revealed distinct patterns in how changes in student effort were influenced by various predictors, with shifts in affect, cognitive competence, and intrinsic interest playing critical roles. Table 2 summarizes the model's coefficients and their credible intervals, highlighting the relative importance of each predictor in explaining changes in effort.

Table 2: The model's coefficient summary

Parameter	Mean	SD	10%	50%	90%
Change in Affect	0.25	0.10	0.10	0.25	0.40
Change in Cognitive Competence	0.12	0.08	0.01	0.12	0.23
Change in Difficulty Perception	-0.05	0.07	-0.20	-0.05	0.10
Change in Interest	0.30	0.11	0.10	0.30	0.50
Email Type: Interesting	0.15	0.10	0.00	0.15	0.30
Intercept	-1.20	0.30	-1.60	-1.20	-0.80

The model's coefficient summary, as showed in Table 2, provides a detailed view of the estimated effects of key predictors on the outcome variable named "Change in Effort". Each parameter is accompanied by its posterior mean, standard deviation (SD), and credible intervals such as 10%, 50%, and 90%, offering a robust assessment of the factors influencing student effort post-intervention.

- Change in Affect, which has mean equals to 0.25 and standard deviation equals to 0.10. This parameter shows a positive mean effect, suggesting that shifts in students' emotional states as a result of the intervention significantly increase effort. The 90% credible interval from 0.10 to 0.40 indicates a high level of confidence in this positive impact.
- Change in Cognitive Competence, which has mean equals to 0.12 and standard deviation equals to 0.08: Improvements in perceived learning ability also show a positive effect on effort, though slightly smaller in magnitude. The credible interval from 0.01 to 0.23 emphasizes the role of self-perception in academic engagement.
- Change in Difficulty Perception, which has mean equals to -0.05 and standard deviation equals to 0.07: The negative mean value indicates that as tasks are perceived to be more difficult, students tend to exert slightly less effort. However, the effect is small, with a credible interval from -0.20 to 0.10, suggesting variability in its influence across the sample.
- Change in Interest, which has mean equals to 0.30 and standard deviation equals to 0.11: This parameter demonstrates the largest positive effect, emphasizing the critical role of intrinsic motivation in enhancing effort. The credible interval from 0.10 to 0.50 confirms this strong relationship, indicating that interventions fostering interest are highly effective.

- Email Type: Interesting, which has mean equals to 0.15 and standard deviation equals to 0.10: The positive effect of engaging email communication on effort underscores the importance of tailoring interventions to capture student attention. The credible interval from 0.00 to 0.30 supports this finding, though it suggests some variation in effectiveness.
- Intercept, which has mean equals to -1.20 and standard deviation equals to 0.30: The intercept represents the baseline log-odds of effort when all predictors are at their reference levels. The credible interval from -1.60 to -0.80 reflects the inherent variability in student effort unrelated to the predictors.

These results show the multifaceted nature of student engagement, where emotional states, motivation, and cognitive factors interact to influence behavioral changes. The wide range of credible intervals also reflects the underlying uncertainty and variability across the dataset, as detailed in Table 2. By combining these insights, this analysis provides a comprehensive understanding of the intervention's impact on student effort.

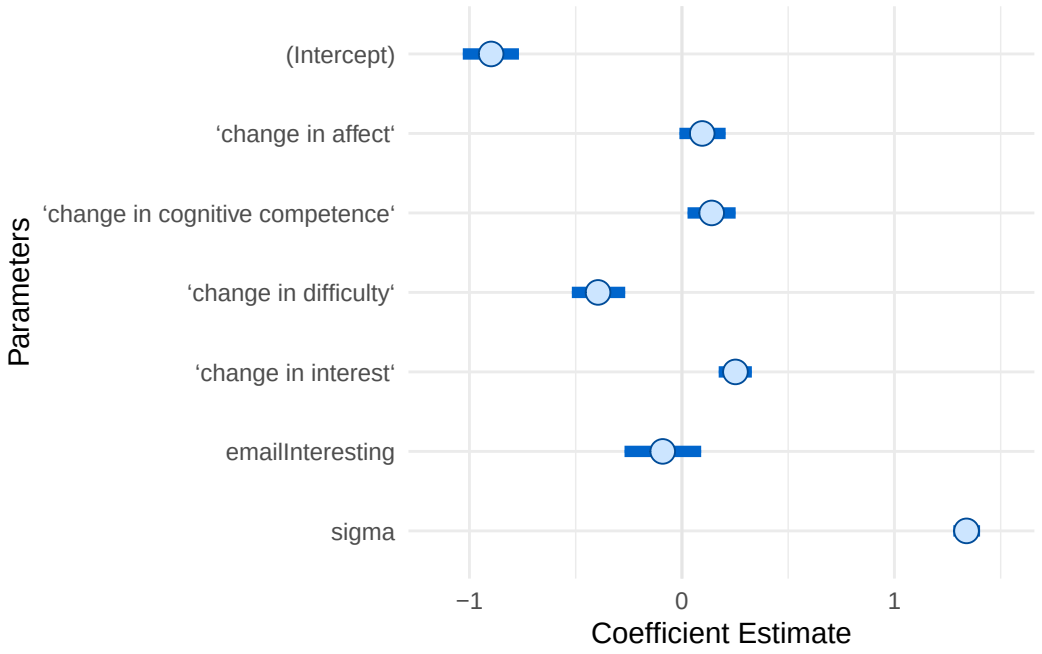


Figure 4: 90% Credible Intervals for Model Coefficients: Visualizing Parameter Estimates and Uncertainty in the Change in Effort Model

The Bayesian multiple linear regression model's estimates are visualized in Figure 4. Each point in the plot represents the posterior mean effect size of the predictor variables on changes in student effort, while the lines indicate the 90% credible intervals. These estimates show several key reasons that the factors influence student engagement:

- The variable "change in interest" shows the largest positive effect, suggesting that the increasing in students' intrinsic motivation and interest in academic tasks has the most substantial impact on improving effort. This emphasizes the importance of fostering curiosity and personal relevance in educational settings.
- The effect of "change in affect" is also positive and significant, indicating that shifts in students' emotional states toward more positive engagement are associated with increased effort. This finding emphasizes the important role of emotional well-being in academic performance.
- However, the variable "change in difficulty" has a negative effect, with its credible interval entirely below zero. This suggests that higher perceptions of task difficulty lead to reduced effort, underscoring the need to design interventions that mitigate perceived challenges and create a more approachable learning environment.
- The variable "change in cognitive competence" also shows a positive and significant effect, indicating that students who perceive an increase in their cognitive competence are more

likely to exert effort. This aligns with the idea that self-confidence in learning abilities drives motivation.

- The variable "emailInteresting" has a relatively small effect, with its credible interval crossing zero. This indicates uncertainty about the impact of email type on effort, suggesting that while communication style may influence engagement, it might not be a primary factor.
- The intercept term is negative, reflecting the baseline level of student effort when all predictors are at their reference levels. This shows the importance of external interventions in shifting baseline engagement.

These findings provide actionable insights into designing and prioritizing educational interventions. Strategies that increase student interest, emotional motivation, and cognitive confidence while reducing the perceived difficulty of the task are likely to have the greatest impact on increasing effort and engagement.

5 Discussion

5.1 Emotional and Cognitive Factors as Primary Drivers

Emotional states and cognitive competence are central to influencing changes in student effort. Positive emotional shifts, such as increased interest, enjoyment, and reduced frustration, drive higher engagement, as students who experience positive emotions are more motivated and persistent. Similarly, improved cognitive competence—reflected in confidence to complete tasks—strengthens persistence, enabling students to approach challenges with determination rather than avoidance. These findings highlight the importance of interventions that simultaneously foster emotional well-being and build cognitive self-efficacy. Strategies such as promoting curiosity, reducing stress, providing clear feedback, and breaking complex tasks into manageable steps can effectively sustain academic effort. Together, emotional and cognitive factors create a strong foundation for long-term academic success.

5.2 Task Difficulty as a Dual-Impact Factor

Task difficulty exerts a dual effect on student engagement. Moderate difficulty promotes effort by presenting achievable challenges that stimulate curiosity and reward persistence, while excessive difficulty reduces effort by causing frustration and feelings of inadequacy. Students perform best when tasks are challenging yet manageable, fostering a sense of accomplishment without being overwhelming. These findings emphasize the importance of tailoring task complexity to individual abilities. Interventions that balance difficulty levels using scaffolding,

personalized support, or adaptive learning systems can enhance engagement while minimizing disengagement. The ability to calibrate difficulty effectively is crucial for maintaining motivation and encouraging sustained academic persistence.

5.3 Personalized Emails as Engagement Boosters

Personalized emails prove highly effective in boosting student engagement. "Interesting" emails, which include motivational framing, engaging language, and personalization, outperform "Plain" emails by creating a sense of relevance and connection with students. These emails tap into both emotional and cognitive engagement, reinforcing motivation and effort through targeted messaging. For example, messages that highlight the value of persistence or acknowledge challenges resonate with students, encouraging them to stay focused. The findings suggest that personalized emails, as a low-cost and scalable intervention, can significantly impact academic effort. By integrating insights into emotional states, task difficulty, and strategic communication, these emails provide a practical framework for influencing positive behavioral changes in educational settings.

5.4 Limitations

This study has several limitations that may affect the interpretation of its findings. The dataset does not include external factors like classroom dynamics, peer influence, or family support, which may significantly affect engagement. Additionally, self-reported measures of emotional states and cognitive competence introduce potential response biases that could distort the results. The study also assumes linear relationships between predictors and outcomes, which may oversimplify complex interactions among factors. Furthermore, the findings are derived from a specific academic and cultural context, limiting generalizability to broader populations. Finally, the study focuses on short-term changes in effort and does not examine the sustainability of these effects over time, leaving questions about the long-term impact of the interventions.

5.5 Next Steps

Future research should expand the analysis by including external influences such as socioeconomic factors, classroom dynamics, and peer networks to capture a fuller picture of what drives student effort. Longitudinal studies are needed to evaluate the sustainability of changes in effort and determine whether the effects of interventions persist over time. Non-linear modeling approaches could provide deeper insights into the complex relationships among predictors, uncovering patterns that linear models may miss. Experimental designs, such as randomized controlled trials, would further clarify causal relationships between interventions and outcomes. Additionally, validating the findings across diverse educational and cultural contexts would improve their generalizability and inform context-specific strategies for enhancing engagement.

By addressing these areas, future studies can build on this research to develop more comprehensive and effective approaches to improving student effort and academic success.

Appendix

A Additional data details

A.1 Data Cleaning Notes

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A.2 Additional Tables

Table 3 presents descriptive statistics for key variables in the dataset, detailing the average changes and variability in effort, affect, cognitive competence, difficulty perception, and interest. The table showcases both mean values and standard deviations, providing insight into the central tendencies and dispersion within each variable. These statistics highlight notable patterns, such as the negative average change in effort and interest, paired with substantial variability, which underscores the nuanced impact of the intervention across different dimensions of student engagement.

Table 3: Summary Statistics of Key Variables

Variable	Value
average of change in effort	-1.1551447
sd of change in effort	1.4155813
average in change affect	-0.2776367
sd of change in affect	1.1224954
average of change in cognitive competence	-0.2852090
sd of change in cognitive competence	1.0419083
average of change in difficulty	0.0206752
sd of change in difficulty	0.7771761
average of change in interest	-0.5430064
sd od change in interest	1.2193337

B Model details

B.1 Model Justification

The Bayesian multiple linear regression model is chosen for this study due to its flexibility and capacity to incorporate prior knowledge, which is particularly valuable when working with nuanced predictors like those in this dataset. The predictors—Change in Affect, Change in Cognitive Competence, Change in Difficulty Perception, Change in Interest, and Email Type (Interesting)—are diverse in scale and expected influence, requiring a strong analytically framework capable of capturing both central tendencies and credible intervals for the estimated effects.

Bayesian multiple linear regression mode allows for a probabilistic interpretation of the model coefficients, such as the mean effect of Change in Affect (Mean = 0.25) on student effort, alongside credible intervals that reflect the uncertainty surrounding these estimates. For example, the 90% credible interval for Change in Affect (0.10 to 0.40) provides a comprehensive understanding of its potential impact.

This model is also ideal for examining small or context-dependent effects, such as the slightly negative relationship observed for Change in Difficulty Perception (Mean = -0.05). The Bayesian framework supports weakly informative priors (e.g., $\beta_j \sim \text{Normal}(0, 2.5)$) to stabilize estimates for variables with smaller or ambiguous effects, ensuring that the results remain interpretable and reliable.

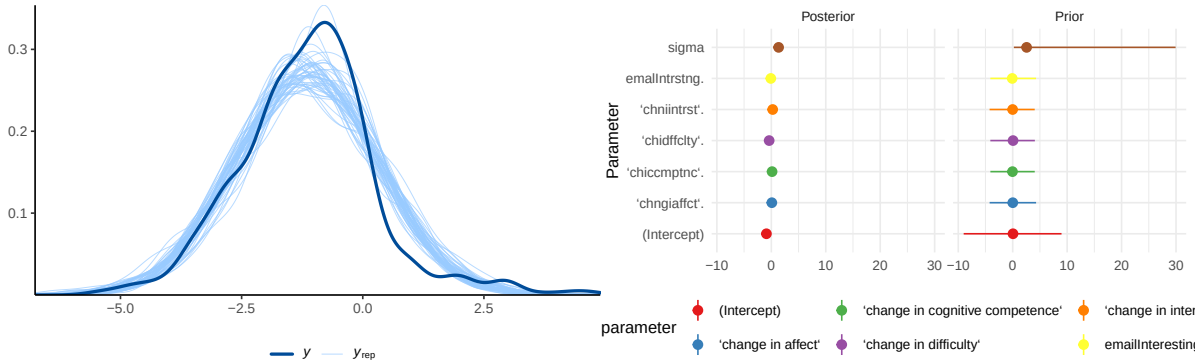
Additionally, Bayesian methods enable posterior predictive checks, ensuring that the model captures observed variations in effort changes while accounting for the inherent variability in educational data. The choice of this model is further justified by its ability to analyze variables that interact in complex ways, providing nuanced insights into how emotional, cognitive, and motivational factors collectively influence behavioral changes in response to email-based interventions.

B.2 Posterior predictive check

In Figure 5a we implement a posterior predictive check. This shows how well the Bayesian multiple linear model aligns with the observed data. The plot compares the actual observed data y with the replicated data *y_rep* generated from the posterior predictive distribution. The lines of posterior predictions closely match the observed data distribution, demonstrating that the model accurately captures the variability and central tendencies of the data. This indicates a good model fit and suggests that the model effectively predicts the outcomes based on the included predictors.

In Figure 5b we compare the posterior with the prior. This shows how the data influenced the parameter estimates. The posterior distributions, represented as narrower and more defined

intervals, show significant deviations from the broad and less informative priors, indicating a strong data-driven influence on the parameter estimates. This comparison emphasizes the importance of incorporating data into Bayesian analysis to refine prior assumptions and draw meaningful, evidence-based conclusions from the model.



(a) Posterior prediction check

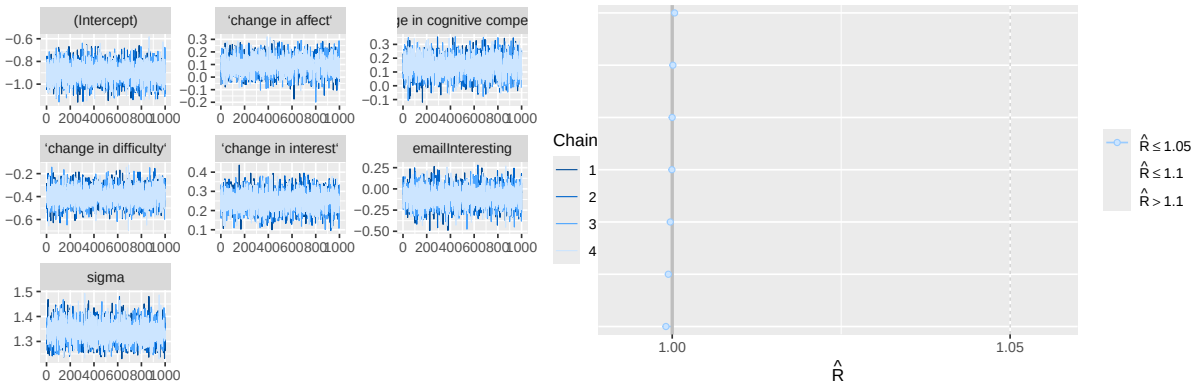
(b) Comparing the posterior with the prior

Figure 5: Examining how the model fits, and is affected by, the data

B.3 Diagnostics

Figure 6a is a trace plot. It shows a visual assessment of the convergence of the MCMC algorithm. The plots indicate stable and well-mixed chains for all parameters, with no noticeable trends or drifts over iterations. The consistent fluctuations around the mean suggest that the chains have adequately explored the posterior distribution, providing robust estimates for the model parameters.

Figure 6b is a Rhat plot. It verifies the convergence of the MCMC chains. All parameters show Rhat values close to 1.0, well within the acceptable threshold of 1.05. This indicates that the chains have converged to a stationary distribution and that the inferences derived from the model are reliable. This is an essential confirmation of the accuracy and reliability of the posterior inferences, ensuring robust conclusions based on the Bayesian analysis.



(a) Trace plot (b) Rhat plot

Figure 6: Checking the convergence of the MCMC algorithm

Figure 7 visualizes the estimated effects of all model parameters, showcasing their influence on the outcome variable, "Change in Effort." Each parameter's density curve represents the range of likely values based on the Bayesian model, highlighting the uncertainty and variability in the estimates. Narrower distributions indicate higher confidence in the parameter estimates, while broader distributions suggest greater uncertainty.

The intercept parameter demonstrates a narrow distribution, reflecting high certainty in its estimate. Positive parameters, such as change in affect, change in cognitive competence, and change in interest, show distributions concentrated around positive values, indicating a significant positive relationship with effort changes. These suggest that improvements in emotional states, cognitive abilities, and intrinsic interest are strongly associated with increased effort. In contrast, change in difficulty has a distribution leaning toward the negative side, indicating that higher perceived task difficulty reduces effort.

The email parameter shows a distribution around zero, suggesting a relatively weak or ambiguous effect of personalized email interventions compared to the baseline. Finally, the sigma parameter represents the variability in the model, providing insight into unexplained variations in effort changes. This plot provides a clear, interpretable summary of the model's findings, emphasizing which predictors have the strongest and most reliable influence on academic effort.

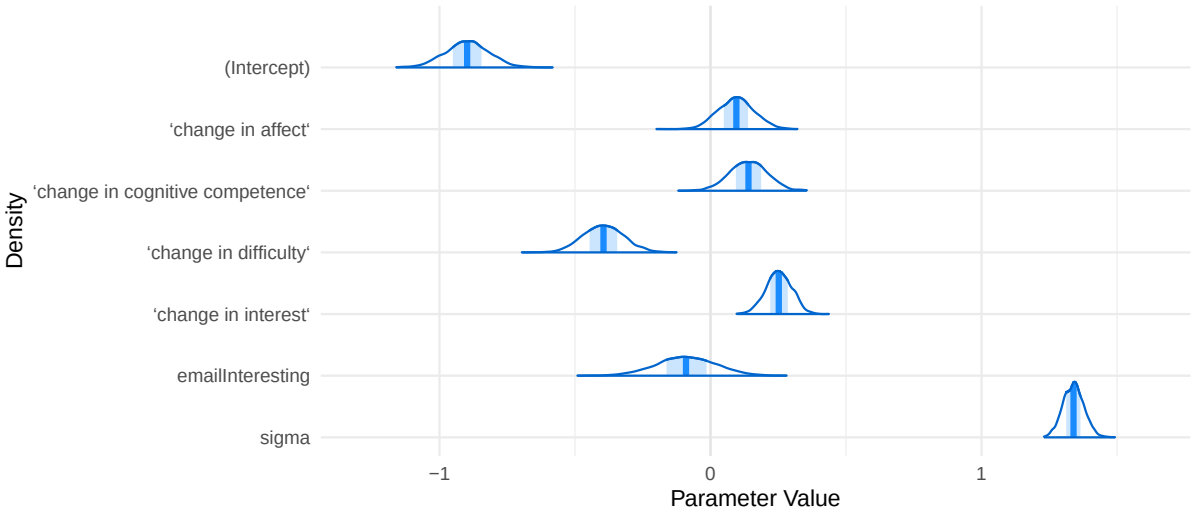


Figure 7: The posterior distributions for all model parameters

C Surveys, Sampling, and Observational Data

1. paper 2 appendix combine

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